

Assignment 6

Exercise 6.1

Part c.

Consider the Spearman's rank correlation test with

H_0 : Crime and Expend Rate are independent,

H_1 : Crime and Expend Rate are dependent,

and the test statistic

$$r_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{[\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2]^{\frac{1}{2}}}$$

where $R_1, \dots, R_n$ and $S_1, \dots, S_n$ are the ranks of Crime and Expend Rate samples, respectively. The statistic r_s is asymptotically normally distributed under H_0 . The `cor.test` function gives $r_s = 0.592$ and the p-value 0.00436. Hence, the test rejects H_0 with the significance level of 5%. This suggests that Crime and Expend Rate are dependent.

Consider the Kendall's rank correlation test with identical test hypotheses as above and the test statistic

$$\tau = \frac{\sum \sum_{i \neq j} \text{sgn}(R_i - R_j) \text{sgn}(S_i - S_j)}{n(n-1)}$$

where $R_1, \dots, R_n$ and $S_1, \dots, S_n$ are the ranks of Crime and Expend Rate samples, respectively. The statistic τ is asymptotically normally distributed under H_0 . The `cor.test` function gives $\tau = 0.429$ and the p-value 0.00476. Hence, the test rejects H_0 with the significance level of 5%. This suggests that Crime and Expend Rate are dependent.

Part d.

Consider permutation test with

H_0 : Crime and Expend Rate are independent,

H_1 : Crime and Expend Rate are dependent,

and the test statistic r_s as stated above in Part c. The function `cor` gives $r_s = 0.592$. The p-value is approximated by the formula given in SDA Syllabus by resampling from the Expend Rate data and evaluating r_s a thousand times. The obtained p-value 0.004 is lower than the significance level 5%. Hence, the test rejects H_0 . This suggests that Crime and Expend Rate are dependent.

Part e.

The modified R code gives that the Kendall's rank correlation test correctly rejects H_0 of 0.50, 0.53, and 0.54 of simulations with sample sizes 48, 50, and 52, respectively. The Spearman's rank correlation test correctly rejects H_0 of 0.49, 0.51, and 0.53 of simulations with sample sizes 48, 50, and 52, respectively. It immediately follows that Kendall's rank correlation test is more efficient than the Spearman's rank correlation test. Notice that the Kendall's test and Spearman's test reject

approximately the same fraction of simulations 0.53 for the sample sizes 50 and 52, respectively. Hence, $are(\tau, r_s) = 52/50 = 1.04$. As $are(\tau, r_s) > 1$, the Kendall's rank correlation test seems to be more powerful than the Spearman's rank correlation test.

Exercise 6.2

Part a.

Consider the Fisher's exact test with

H_0 : Gender and Fatalities among Infected are independent,

H_1 : Gender and Fatalities among Infected are dependent,

and the test statistic N_{11} which is the cell frequency of the cell (1,1) in the provided 2x2 contingency table. The statistic N_{11} follows $Hyp(2234, 1097, 47)$ under H_0 . The function `fisher.test` gives the p-value of 0.054, which means that the test does not reject H_0 with the significance level of 5%. Hence, Gender and Fatalities among Infected may or may not be dependent.

Part b.

Consider the Fisher's exact test with

H_0 : Death from infection is not more common among men than women,

H_1 : Death from infection is more common among men than women,

and the test statistic N_{11} as stated above in Part a. The function `fisher.test` gives the p-value of 0.029, which means that the test rejects H_0 with the significance level of 5%. This suggests that deaths from infection are more common among men than women.

Part c.

The function `phyper` gives the left p-value 0.99 and the right p-value 0.029 which coincides with the p-value obtained in Part b. above.

Exercise 6.3

Part a.

We find model II B most suitable for this data because it is expected that the rows are independent from each other, and we know the sample sizes.

Part b.

5% is used as significance level and use model II B.

H_0 : The 5 samples are homogeneous.

H_1 : The 5 samples are not homogeneous.

The Chi-Square test shows we can reject the null hypothesis, so we show that the samples are not homogeneous. We can do that because the p-value is 0.06952 (> 0.05) and in the expected values nothing stands out. See table below.

	Number of Patients	Incidence of Nausea
Placebo	174.59394	85.40606
Chlorpromazine	136.98909	67.01091
Dimenhydrinate	91.99758	45.00242
Pentobarbital(100mg)	68.49455	33.50545
Pentobarbital(150mg)	81.92485	40.07515

Figure 1: Excepted values of Chi-Square test.

Part c.

A Chi-Square test with simulate on results in a p-value of 0.07846. This number is close enough to zero the say the results are ridable.

Part d.

	Number of Patients	Incidence of Nausea
Placebo	-0.7260765	1.0381321
Chlorpromazine	1.2825196	-1.8337251
Dimenhydrinate	-0.7295573	1.0431089
Pentobarbital(100mg)	-0.1805848	0.2581971
Pentobarbital(150mg)	0.3397494	-0.4857680

Figure 2: Contributions Chi-Square test

	Number of Patients	Incidence of Nausea
Placebo	-1.53	1.53
Chlorpromazine	2.58	-2.58
Dimenhydrinate	-1.39	1.39
Pentobarbital(100mg)	-0.34	0.34
Pentobarbital(150mg)	0.64	-0.64

Figure 3: Standard Residuals Chi-Square test

The categories that stand out after the residual values and standardized values (5% = 1,96) is the Chlorpromazine. The rest is in line with the rejection of the H0.

Part e.

Bootstrap 1000 time that resulted in a p-value of 0.059. So, the results are in line with the finding from Part b.

Part f.

To test if the placebo works better than the Chlorpromazine, a fisher test is used.

H0: The placebo works better than the Chlorpromazine.

H1: The placebo works not better than the Chlorpromazine.

The p-value is 0.007148 so we reject the H1. The Chlorpromazine is better than the placebo.

Appendix

```
source('functions_Ch3.txt')
source('functions_Ch5.txt')
source('functions_Ch7.txt')
source('expensescrime.txt')
source('nausea.txt')
library(mvtnorm)
```

Exercise 6.1

Part c.

```
data61 <- read.table("expensescrime.txt", header=T)
expend_rate = (data61$expend/data61$pop)[data61$pop >= 4000]
crime = data61$crime[data61$pop >= 4000]
cor.test(expend_rate, crime, method = "k")
cor.test(expend_rate, crime, method = "s")
```

Part d.

```
B = 1000
t = cor(expend_rate, crime, method="s")
t_perm = replicate(B, cor(expend_rate, sample(crime), method="s"))
p = 2 * min(sum(t_perm >= t)/B, sum(t_perm <= t)/B)
```

Part e.

```
aresimulation=function(B,n){
  pvalken = pvalspear = numeric(B)
  for(i in 1:B){
    x = rmvt(n,sigma=matrix(c(1,0.3,0.3,1),2,2),df=6)
    pvalken[i] = cor.test(x[,1], x[,2], method = "k")[[3]]
    pvalspear[i] = cor.test(x[,1], x[,2], method = "s")[[3]]
  }
  powerken = mean(pvalken<0.05)
  powerspear = mean(pvalspear<0.05)
  rbind(c("ken", "spear"),c(powerken,powerspear))
}
```

```
}  
aresimulation(5000,48)  
aresimulation(5000,50)  
aresimulation(5000,52)
```

Exercise 6.2

Part a.

```
m = matrix(c(30, 17, 1067, 1120), nrow = 2, ncol = 2)  
fisher.test(m)
```

Part b.

```
fisher.test(m, alternative = "g")
```

Part c.

```
pl = phyper(30,1097,1137,47)  
pr = 1 - phyper(30-1,1097,1137,47)
```

Exercise 6.3

```
data63 = read.table("nausea.txt", header = T)
```

#C because the null hypothesis is that the data points are onafhankelijk.

Part b.

```
data63  
chisq.test(data63)  
chisq.test(data63)$expected  
chisq.test(data63, correct= F)
```

Part c.

```
chisq.test(data63, simulate.p.value = T)
```

Part d.

```
chisq.test(data63)$residuals  
round(chisq.test(data63)$stdres,2)  
#alpha = 5% so 1,96
```

Part e.

```
data63  
max <- maxcontributionscat(data63)  
bb <- (bootstrapcat(data63, 1000,maxcontributionscat))  
mean(bb>=max)
```

Part f.

```
matrixa <- data63[1:2,]  
matrixb <- matrix(c(165,95,152,52), nrow = 2, ncol = 2, byrow = T)  
matrixb  
fisher.test(matrixb, alternative = 'l')
```